

Distributed Classical–Quantum Algorithms for Certifiable Trustworthy Machine Learning

Overview

Modern machine learning is increasingly used in high-stakes decisions, yet many training methods remain heuristic, opaque, and difficult to certify. Although interpretable models such as decision trees, clustering models, and policy trees improve transparency, training them to certified optimality remains challenging due to large-scale, nonconvex, and discrete optimization structures. My long term goal is to develop a new scalable algorithmic foundation for certifiable machine learning through distributed classical–quantum optimization. The goal of this proposal is to train scalable trustworthy learning models with provable feasibility, optimality, or near-optimality certificates, rather than relying only on gradient or heuristic search. The research will be organized around four integrated thrusts. i) develop reduced-space formulations and certificate-generating algorithms for general learning problems. ii) design distributed decomposition methods that split large learning problems into smaller subproblems while maintaining global consistency and certificate tracking. iii) create quantum-assisted and quantum-inspired optimization modules for selected decomposed subproblems. iv) integrate neural guidance with certificate-preserving global optimization to accelerate search without sacrificing reliability.

The education plan is tightly integrated with the research vision. The project will create a new educational pathway on certifiable and quantum-enabled trustworthy AI. This will include new course modules in deep learning analytics and optimization-based machine learning, undergraduate research projects on interpretable and certifiable AI, and open-source teaching materials that visualize branch-and-bound, quantum (-inspired) based question reformulation and optimization, and optimality-gap certificates. The project will also develop age-appropriate K–12 outreach activities that introduce core ideas such as interpretable AI, decision trees, and optimization through interactive demonstrations. The project will broaden participation through above actions in the intersection of AI and future computing.

Intellectual Merit

The intellectual merit of this project lies in advancing **certifiable machine learning**, where trustworthy models are trained for predictive performance, interpretability, and rigorous guarantees on feasibility and solution quality. The project will extend deterministic machine learning to address real-world needs in scalability, decomposition, hardware acceleration, and reliable certification. Its innovations include i) identifying structural conditions for compact search spaces and certifiable training algorithms; ii) developing distributed optimization methods that decompose large learning problems while preserving feasibility, global consistency, and optimality-gap information; iii) incorporating quantum-assisted and quantum-inspired solvers as certificate-preserving accelerators; and iv) designing neural-guided branching, warm starts, and variable fixing without compromising certification. Together, these advances will move trustworthy machine learning beyond heuristic training toward scalable, interpretable, and certificate-aware optimization.

Broader Impacts

The broader impacts of this project are to advance trustworthy AI, train the next-generation computing workforce, and broaden access to interdisciplinary research at the intersection of machine learning, optimization, and quantum computing. By developing interpretable learning algorithms with explicit feasibility and solution-quality certificates, the project will support more transparent, auditable, and reliable AI systems for high-stakes domains such as healthcare, transportation, manufacturing, and energy. The proposed research will serve as the dissertation topic for two Ph.D. students, one of whom will be from an underrepresented group in STEM; both students will be mentored to publish in scientific journals and present their work at international conferences.